PHYSICAL INTELLIGENCE FOR JAPAN'S MANUFACTURING FUTURE

In Situ Certification of Autonomous Welds: A Measurement and Evidence Layer for High-Consequence Manufacturing, from Niche Applications to Japan's Strategic Industries

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Budget: ¥600M to ¥1.2B (about \$4M to \$8M USD) over 36 months

Executive Summary

Autonomous welding is no longer the hard problem. Japan's major integrators have already moved on the robot and the simulation layer; the capability gap that now blocks deployment is certification: the ability to prove that a weld no human supervised is fit for service. This program builds that missing layer, an in situ certification capability that establishes, from measurements taken as the weld forms, the evidence a certifying authority needs to accept an autonomous weld without post-weld inspection.

The need is most critical in harsh environments and in cases with limited access. In fusion reactor construction, spacecraft, and orbital and lunar structures, no inspector will be able to access them, and the welds are one-off, high-consequence, and often made of materials and microstructures generated on the fly that were never qualified beforehand. There is no prior specification to inspect against, and conventional post-weld nondestructive testing (NDT) cannot run in these environments at all. These developments apply directly both in these difficult conditions and across Japan's strategic industries, including shipbuilding, nuclear, automotive, and heavy construction, where the certified welding and inspection workforces are retiring in parallel and the shortage must be addressed by new technologies such as autonomous welding, in situ monitoring, automated inspection, and new pathways to certification.

The program rests on three elements: 1) a statistical framework for certification evidence that defines what in situ measurement must show to substitute for post-weld NDT. 2) a multimodal sensing platform that captures internal, surface, and microstructural state during deposition. As an example, laser ultrasonics is a technique that can interrogate the welding process in situ, is non-contact, and can operate in the high-temperature environment, where X-ray and conventional ultrasonic testing cannot. 3) a multinational team spanning Japan and the United States, with at least half the work performed in Japan. The ask is ¥600M to ¥1.2B over 36 months, roughly 0.2% of METI's FY2026 physical AI foundation model program, and the layer that determines whether that program's output can ever be deployed in a regulated weld.

The Challenge

Every welded structure in Japan's strategic infrastructure follows the same procedure: a weld is made, then inspected (visually, ultrasonically, or radiographically), and only then certified for service. Skilled welders execute the first step, certified inspectors the second, and both are required for the third. Automation has now largely solved the first step; what it cannot yet do is the second and third, namely certify its own output. Every robot installed in a Japanese fabrication shop still depends on a human inspector to accept the result.

Situations with limited or no access for human beings such as fusion reactor construction, spacecraft, orbital and lunar structures, nuclear power plants, and confined offshore joints are precisely the environments in which post-weld inspection is impossible or hazardous, the consequence of failure is highest, and the welds are one-off rather than mass-produced. NASA is already funding this gap directly: recent Early Stage Innovations awards support work such as weldability assessment for conditions in space using a digital twin, on the explicit premise that the physics of welding is well understood on Earth but not in space, and that astronauts cannot easily inspect or repair what they build. These markets are small today, but they are real, since no incumbent has certified an autonomous weld where no human can verify it, and they are among the hardest cases, which makes them one strong setting in which to prove the capability.

The same shortage then compounds across the sectors Japan has named strategic. Welding is foundational to the 2040 industrial agenda: automotive, shipbuilding, construction machinery, semiconductor equipment, energy systems, and seismic retrofitting all rest on welded joints. And the shortage is uniquely doubled, because each retiring certified welder is paired with a retiring certified inspector. Unlike most skilled trades, welding fails catastrophically rather than gracefully, which makes the downside a national resilience risk, not service degradation.

The deeper reason certification is the binding constraint is that the ground is shifting beneath it. Traditional certification works from a predicate: a weld procedure is qualified in advance against a known material and a fixed acceptance specification, and the finished weld is inspected for conformance. Eric Schmidt and Dror Berman describe software that agents now generate "on the fly," adapted per context and never read by a human. Ayman Salem has named the physical analog directly: robotic metal deposition in which an AI agent, drawing on materials informatics, produces alloys, thermal histories, and microstructures specific to each location that no engineer specified or qualified beforehand. When the artifact is generated this way, there is no prior specification to inspect against. Conformance testing after the weld is then not merely slow or constrained by labor; it is conceptually impossible, and the only thing that can be certified is the material as it forms. (Schmidt & Berman, "When Machines Build for Machines," 2026)

The Opportunity

Three forces make this the moment to act. First, the sensing is real, not hypothetical. Published research in Japan has demonstrated laser ultrasonic detection of internal weld defects (lack of penetration, solidification cracking, incomplete fusion, blowholes, and melt width) in situ during welding, taken it onto a production steam turbine rotor, and integrated it onto an arc welding robot using a compact microchip laser, detecting blowholes at production speed. Laser ultrasonics is a strong candidate because it is non-contact, remote, broadband, and works in the high-temperature field during deposition, reading internal state where optical and thermal sensors see only the surface and where X-ray cannot run during welding. Machine learning inference in real time on weld signals has separately been demonstrated, so the pieces for an in situ record of material state, inferred by AI, now exist.

Second, the incumbents have moved on everything except this layer. Japan's leading integrators have committed decisively to the robotics and simulation layer. Kawasaki Heavy Industries opened a Silicon Valley physical AI center with NVIDIA and partners in May 2026, and Hitachi has built on an NVIDIA partnership since 2024, but neither is aimed at the in situ measurement and certification of the weld itself. The robot and the simulation are being solved; the layer that decides whether an autonomous weld can be accepted for service is open.

Third, the policy environment is aligned and the certification pathway is opening. METI's FY2026 program for AI robot and physical AI foundation models (¥387.3 billion) explicitly targets this intersection, with a national goal of roughly 30% of a ¥60 trillion global physical AI market by 2040. But that program funds the model; nothing in it funds the certification layer that lets the model's output be deployed in a regulated weld. Meanwhile JSNDI, as secretariat of ISO/TC 135, is active in the global NDE 4.0 digitalization agenda, and ASTM has begun publishing standards for registering in-process monitoring data as a basis for qualification, institutional openings that did not exist five years ago.

The Approach

The program builds one capability, an in situ certification layer for autonomous welds, and proves it across two generalized cases: harsh environments or locations with limited accessibility, where qualification in advance is impossible and no inspector can access the weld, and scale, where the same platform relieves the labor shortage in Japan's strategic industries. It rests on three components.

1. The Certification Evidence Framework

The central question is not how to build a better sensor or a better robot; it is what evidence a certifying authority must have to accept in situ measurement in place of post-weld NDT. This is the program's fundamental challenge and its principal investigator's specialty. The framework generalizes the tradition of reliability statistics used to qualify inspection systems, including probability of detection (POD) curves, ROC-based qualification, and Bayesian inference on small samples, extending it from a single post-weld measurement to a continuous evidence stream that runs across modalities during the process. Its hardest problem, and its core research contribution, is transferability: establishing when evidence earned on one material, joint geometry, or machine carries to another, since autonomous certification cannot be derived again for every weld. Certification authorities are engaged from the outset to help define what the evidence record must contain, so the output is not a research artifact but a qualification basis that code committees can adopt. In effect, this is the evaluation layer for a world Schmidt and Berman describe but do not resolve, one in which we build systems whose outputs exceed our ability to evaluate them. The contribution is to make a weld that was never qualified beforehand, and need not be fully understood, demonstrably and quantifiably fit for service.

2. The Multimodal Sensing Platform

Certifying an autonomous weld requires more than one signal. Borrowing the logic a certifying inspector already applies, which relies on independent, redundant evidence rather than a single indicator, the platform fuses three orthogonal modalities into one inferred record of material state: laser ultrasonics for internal and volumetric state (defects, penetration, melt geometry); thermal, optical, acoustic, and high-temperature eddy current sensing for surface and process state (thermal history, cooling rate, bead geometry); and microstructure informatics to link that measured thermal history to the microstructure and mechanical properties that ultimately define fitness for service. Physics-informed machine learning fuses the streams, and the evidence framework qualifies the result. The same instrumented platform that validates certification during the process also generates the labeled ground truth needed to qualify each modality, so the two are outputs of one investment, not separate programs.

3. Niche Applications and Scale

In harsh and/or inaccessible conditions (space, lunar and orbital structures, nuclear power plants) the platform's non-contact, remote, in situ character is not a convenience but the only option, and because the joints are generated on the fly and were never qualified in advance, in situ certification is the only paradigm that can apply. The program does not presume which setting comes first; the entry point is chosen on the basis of validation evidence, transition-partner readiness, and the certification pathway. The same capability applies directly across Japan's strategic industries, including shipbuilding, automotive, heavy construction, and additive repair, where the same record lets one experienced inspector oversee many autonomous welds and addresses the labor shortage regardless of how quickly formal code reform proceeds. Even where post-weld NDT remains mandated, the platform reduces the burden per weld on a shrinking inspector workforce, so the program delivers value across the full transition, not only at its endpoint.

Program Metrics and Milestones

Progress is tracked in three categories adapted for certification: speed, accuracy, and transferability. Speed is in-process latency, the certification-relevant evaluation delivered while the weld region is still actionable and keeping pace with deposition rather than as an offline computation. Accuracy is certification-decision quality with the rigor of modern statistical methods for acceptance criteria. Transferability, the program's core research problem, is measured by demonstrating the framework across at least two materials with distinct microstructures, at least two welding cells, and multiple joint configurations, with independent reproduction at a second site. Targets are set from the acceptance criteria of the selected application rather than imported, and every metric is demonstrated across a deliberately wide range, from proper welds to welds with intentional defects.

This is a 36-month effort: integrate multimodal in-process sensing (about month 5); convert the sensed signal into quantified internal-state evidence conditioned by a physics-based microstructure model (about month 11); operate the certification evidence framework on the continuous in-process stream (about month 16); validate against reference characterization and demonstrate initial transferability across two materials and two cells (about month 22); demonstrate a pilot in a representative high-consequence setting (about month 28);

reproduce the workflow at an independent site (about month 31); and prepare standards input, with the method and detection-reliability framework directed to ISO/TC 135 and the weld-acceptance side to the relevant welding and product committees (about months 33 to 36).

Why me?

Jeremy S. Knopp, Ph.D., brings the rare combination this program requires: the statistics to define the evidence and the standing to convene the institutions. His background spans materials science, reliability statistics, and 25+ years in research and R&D management, including service as Technical Director at AFOSR Tokyo (AOARD), where he managed large international R&D portfolios at the intersection of advanced manufacturing and dual-use technology. His M.S. in statistics, focused on the reliability of industrial inspection systems and probability of detection, is the methodological foundation of the evidence framework, and he served as technical monitor for the revision of MIL-HDBK-1823A, the enduring standard for POD studies. He is an active member of JSNDI and advises dual-use startups in the U.S. and Japan, and is precisely the bridge, between the statisticians and the engineers and between Japan and the United States, that this problem has historically lacked.

The program is structured to convene a multinational set of demonstrated capabilities, with the majority of the work performed in Japan leveraging international collaborations as required for program success. On the Japan side, leading Japanese welding and NDT research institutes are candidates to anchor the internal sensing layer, building on laser ultrasonic systems already demonstrated in production at the Joining and Welding Research Institute (JWRI) at Osaka University, and Japan's heavy industry integrators provide the path to industrial scale. This is not a startup seeking market entry. It is a research program with institutional standing within Renaissance Philanthropy's BiTS Japan Cohort and the Global Startup Campus (GSC) Initiative, with access to Japan Cabinet Office and METI channels, led by a principal investigator who already works inside the certification community whose acceptance the program requires.

Budget Overview

- Year 1 (¥200M to ¥350M): Instrument two pilot welding cells (one in Japan, one in the U.S.) with the multimodal sensing platform; baseline laser ultrasonic and process sensor data collection; first physics-informed inference models; initiation of a JSNDI working group on the in situ evidence standard; definition of the evidence framework and the transferability problem with certifying authorities.
- Year 2 (¥200M to ¥450M): Closed-loop certification demonstrations under production conditions; multimodal fusion and linkage between microstructure and properties; Statistical qualification trials, including transferability across materials and machines; preliminary code committee engagement with supporting documentation; scoping of a pilot in either an access-limited setting (such as space or nuclear, including fusion) or a strategic-industry setting (such as shipbuilding).
- Year 3 (¥200M to ¥400M): Deployment across multiple sites; an in situ certification pilot in the selected setting, whether one where post-weld inspection is impossible or a strategic-industry weld where it is a bottleneck; documentation of the regulatory pathway and an open data release to support code revision; transfer of the qualified evidence framework to JSNDI and ASTM for standards integration.

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